

What is Next.js?

- Next.js is a React framework for building full-stack web applications.
- You use React Components to build user interfaces.
- Next.js adds features and optimizations on top of React.

- It automatically configures lower-level tools like bundlers and compilers.
- Developers can focus on building and shipping products quickly.
- Works well for individual developers and larger teams.
- Helps build interactive, dynamic, and fast React applications.

Getting Started with Next.js

Installation

System Requirements

- Node.js 18.18 or later
- macOS, Windows (WSL), or Linux

Automatic Installation

```
npx create-next-app@latest
```

Installation Prompts

You'll be asked:

- Project name
- TypeScript (Yes/No)
- ESLint (Yes/No)
- Tailwind CSS (Yes/No)
- Use src/ directory (Yes/No)
- Use App Router (Yes/No)
- Use Turbopack (Yes/No)
- Import alias setup

Manual Installation

Install packages manually:

```
npm i next react react-dom
```

App Directory

- **app/layout.tsx** → Root layout (must include **<html>** + **<body>**)
- **app/page.tsx** → Home page

Public Folder

- **Optional: Add static files (images, fonts).**
- **Example:**
 /public/profile.png → Access
 as/profile.png

Run Dev Server

Start your project:

```
npm run dev
```

Visit <http://localhost:3000>

TypeScript Setup

- Rename files to .ts / .tsx
- Run next dev → auto-installs dependencies
- Generates tsconfig.json

Path Aliases

Use absolute imports with tsconfig.json or jsconfig.json:

```
// Before  
import { Button } from  
'../../components/button'
```

```
// After  
import { Button } from  
'@/components/button'
```